

Practice problems #4

1. True or False. Justify:

- (a) If a matrix is in reduced echelon form, then it is in echelon form.
- (b) If a 3×4 coefficient matrix has no leading entry in the 3rd row, then there is no solution for some right hand side.

2. For the linear system

$$\begin{cases} x_1 - 3x_2 = 1 \\ 2x_1 + hx_2 = k \end{cases}$$

give the augmented matrix, and apply row operations as we did in class to answer the following:

Choose h and k such that the system has (a) no solution, (b) a unique solution, or (c) infinitely many solutions. Give separate answers for each part.

3. Give the solution set of a linear system whose augmented matrix is row equivalent to the following matrix

$$\begin{bmatrix} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

by setting $x_3 = s$, and $x_4 = t$.

Answers or Hints are on page 2.

#1(a): T. Why?

#1(b): T. Why?

#2(a): $h = -6$, and $k \neq 2$ (Note that you need $0 = \text{nonzero}$.)

#2(b): $h \neq -6$ (Note that you need leading entries in every column except the last column).

#2(c): $h = -6$, and $k = 2$ (Note that you need $0 = 0$.)

#3: $\text{Sol} = \{(3 + 5s, 6 - 4s + t, s, t, 0) : s, t \in \mathbf{R}\}$